

AQUASENSE

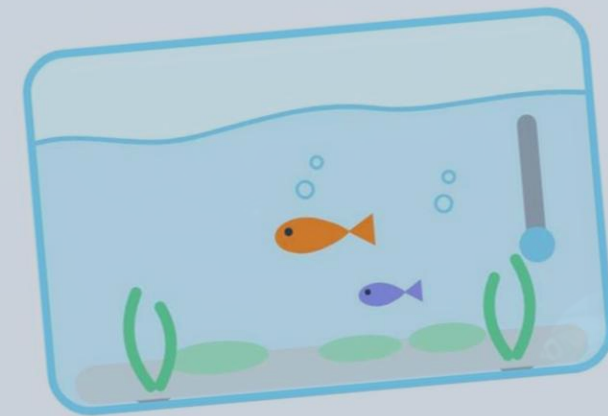
Real-time aquarium monitoring system, built for safer aquatic ecosystems.



AQUASENSE

Welcome Aboard!

Set up your smart aquarium monitor in three simple steps :)



- 1 Set up your aquarium**
Name your aquarium and choose the template its safe ranges come from or create a custom one.
- 2 Mount your hub**
Mount the hub and attach the four AquaSense sensors to your aquarium.

Get Started

THE PROBLEM

Aquarium ecosystems

Manual monitoring

Water parameters are difficult and time-consuming to test consistently.

Environmental Changes

Changes to the aquarium ecosystem could put marine life health at risk

Complex Care

Disconnected tools create blind spots instead of confidence.



Our Mission

“Providing aquarium owners an aquarium monitoring system to better protect and build safer environments for their marine animals”

STAKEHOLDERS

Who we do this for

- ✓ Hobbyists
 - Casual aquarium owners looking to take the next step in improving their aquarium maintenance
- ✓ Business owners
 - Balancing multiple aquariums means having a better automated system to track quality

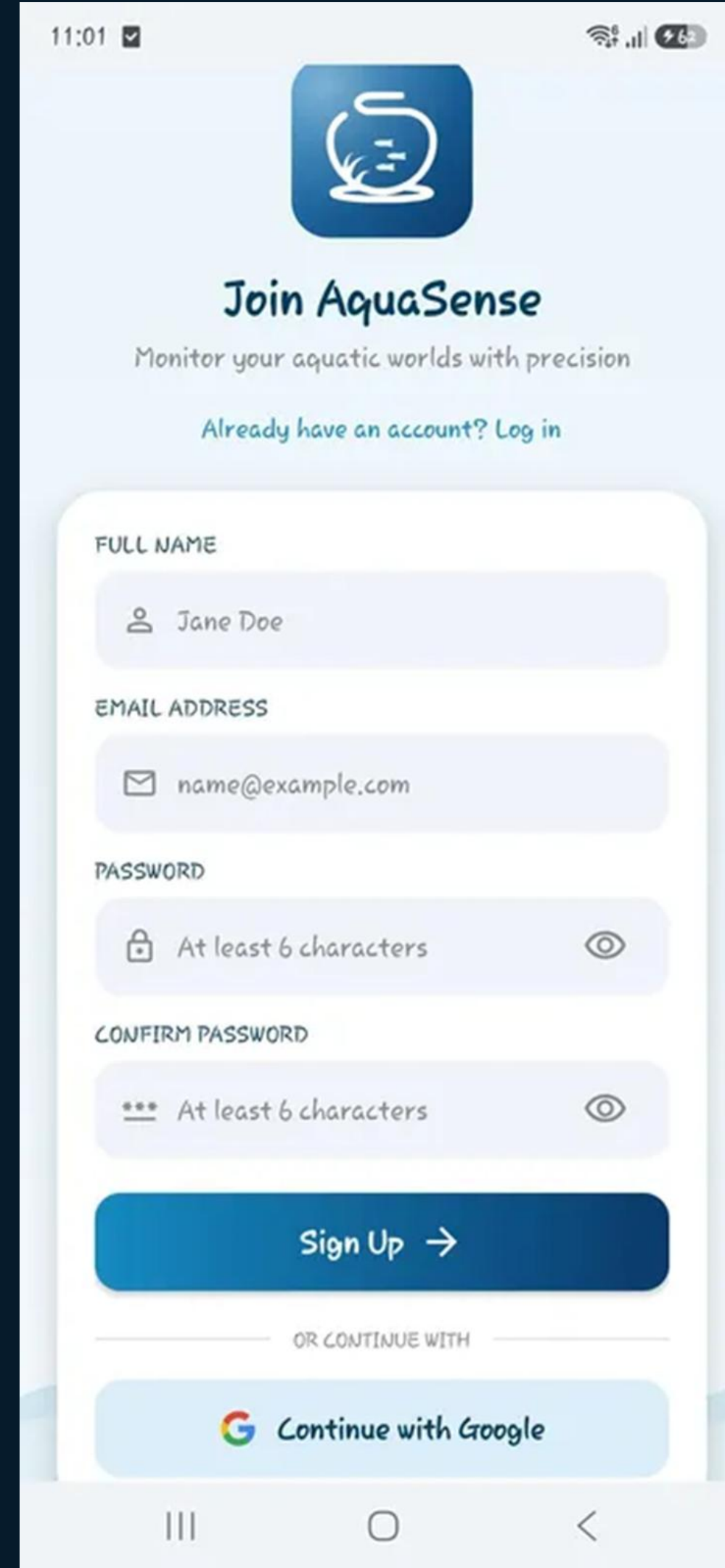


OUR SOLUTION: AQUASENSE

AUTOMATED MONITORING

- pH
- Temperature
- Total Dissolved Solids (TDS)
- Water level

One app for multiple aquariums, instant alerts and historical analytics.

A screenshot of a mobile app's sign-up screen. At the top, the status bar shows the time 11:01, a checkmark icon, and signal/battery indicators. The app's logo, a blue square with a white fishbowl icon, is centered at the top. Below it, the text "Join AquaSense" is displayed in a bold, dark font, followed by the tagline "Monitor your aquatic worlds with precision" in a smaller, lighter font. A link "Already have an account? Log in" is positioned below the tagline. The main form area contains four input fields: "FULL NAME" with a person icon and the text "Jane Doe", "EMAIL ADDRESS" with an envelope icon and the text "name@example.com", "PASSWORD" with a lock icon, the text "At least 6 characters", and an eye icon, and "CONFIRM PASSWORD" with three asterisks, the text "At least 6 characters", and an eye icon. A large blue button labeled "Sign Up →" is located below the password fields. Below this button, the text "OR CONTINUE WITH" is centered, followed by a light blue button featuring the Google logo and the text "Continue with Google". The bottom of the screen shows the standard Android navigation bar with three icons: a square, a circle, and a triangle.

GOALS & OBJECTIVES

01 PROTECT

Alerting users regarding the safety of their pets.

02 SIMPLIFY

Tracking water parameters made easy to understand.

03 CONNECT

Connecting the hardware to database to mobile app.

04 ANALYSE

Store and analyze data to make further aquarium improvements.

Key Features of our Product

Made by and for aquarium hobbyists, a packaged, attachable sensor system that sends real-time data to the AquaSense app.

LIVE READINGS

Real-time monitoring of critical water parameters sent directly to the app

AQUARIUMS

Manage multiple aquariums in ONE app

ALERT

Instant alert notifications for critical conditions

ANALYTICS

Graphs all water parameter trends for users to analyse

EASY INSTALLATION → USER-FRIENDLY INTERFACE

Product Overview

SENSORS

PH, TDS, Temperature and Water Level.

UNIVERSAL

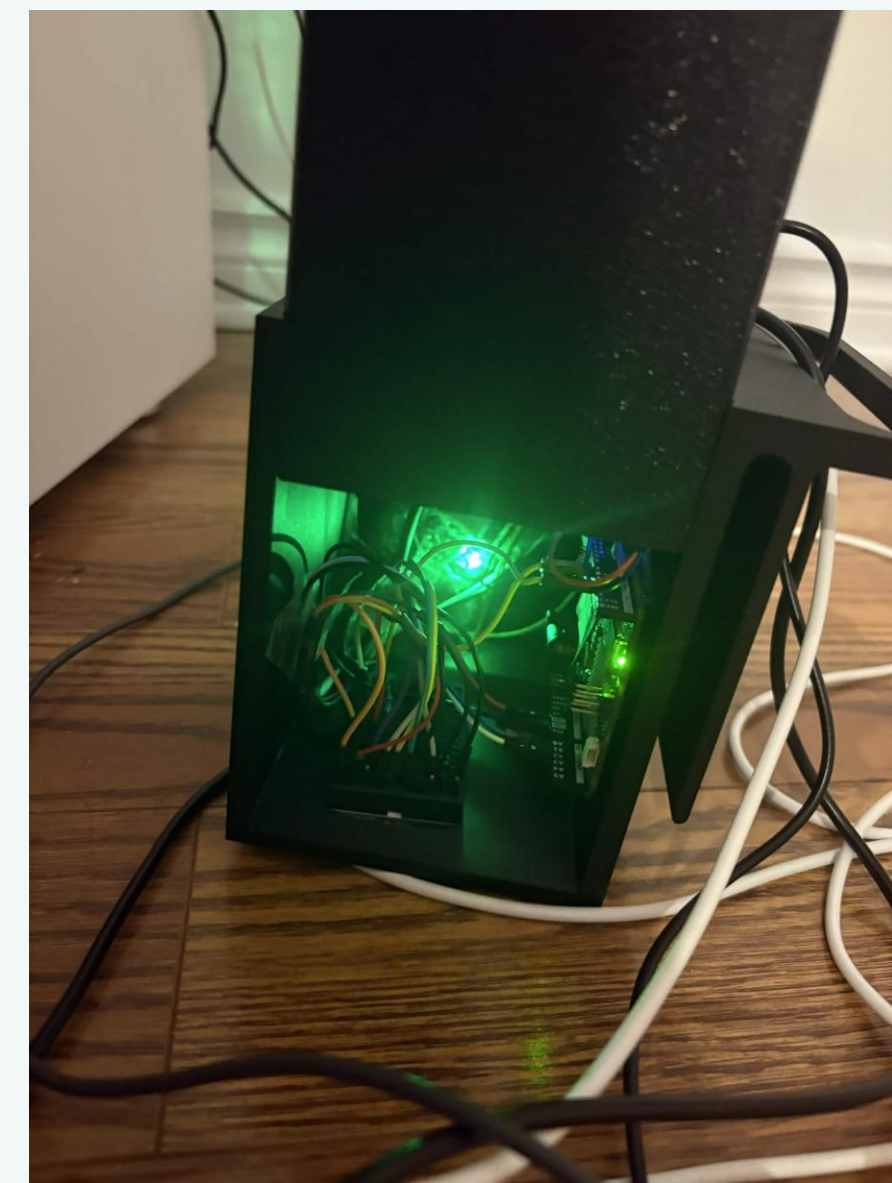
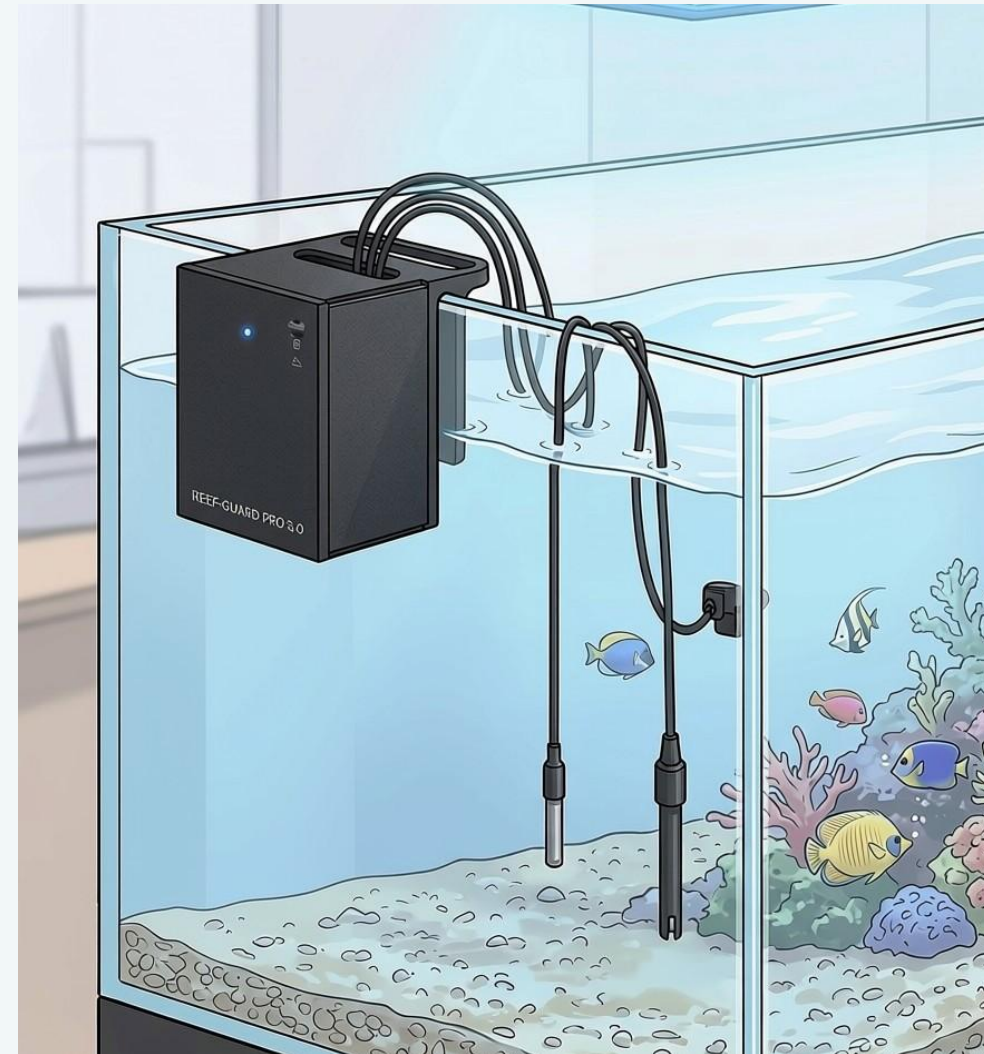
Sensors were selected to match the basic needs of all types of aquariums.

CONNECTED

Arduino Uno R4 with built-in WiFi

CLOUD-SAFE

Sensor data stored in the cloud to prevent data loss



How It Works

SENSE

The four probes take continuous pH, TDS, temperature and water-level readings.

READ

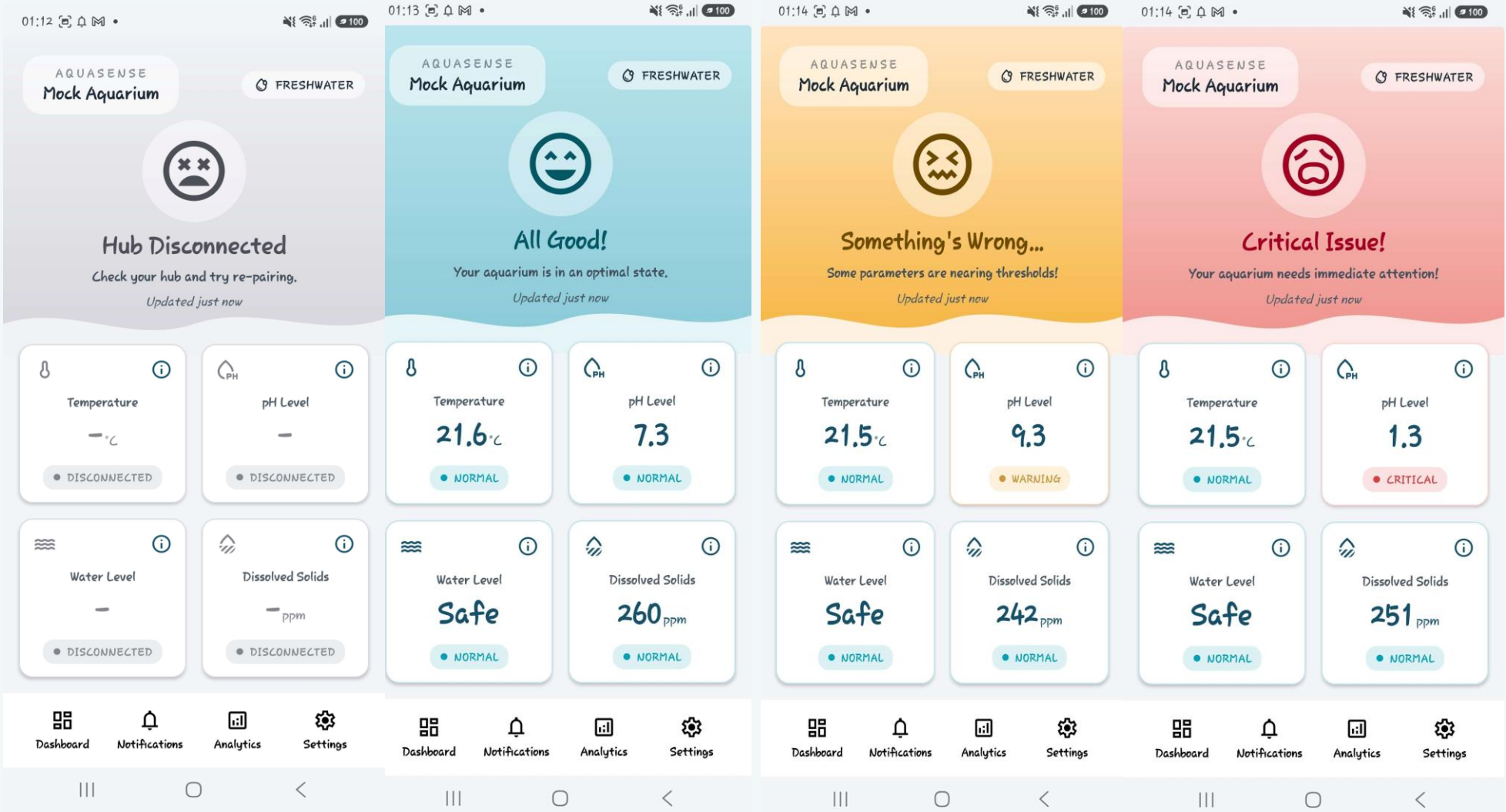
The Arduino Uno R4 digitizes each reading and connects over built-in WiFi.

STORE

Readings are pushed to the Firebase cloud database, timestamped and saved.

SHOW

The Android app reads live values, displays them, and fires threshold alerts.

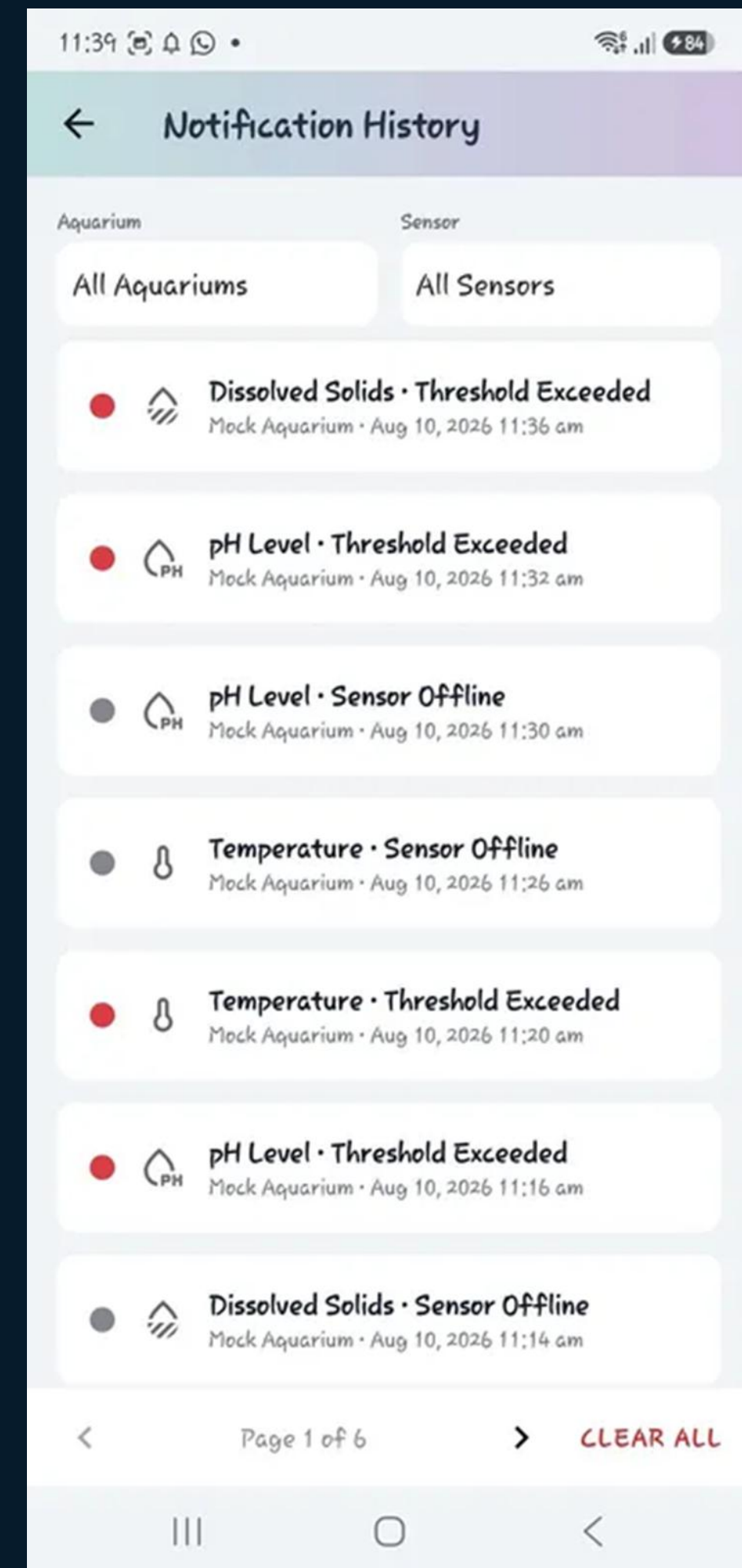


App Benefits: Notifications

From data collection to alerts

Aquariums depend on many interacting factors. AquaSense automates collection, delivers readings directly to users, and triggers alerts when conditions become critical.

CRITICAL CHANGE → NOTIFIED IMMEDIATELY

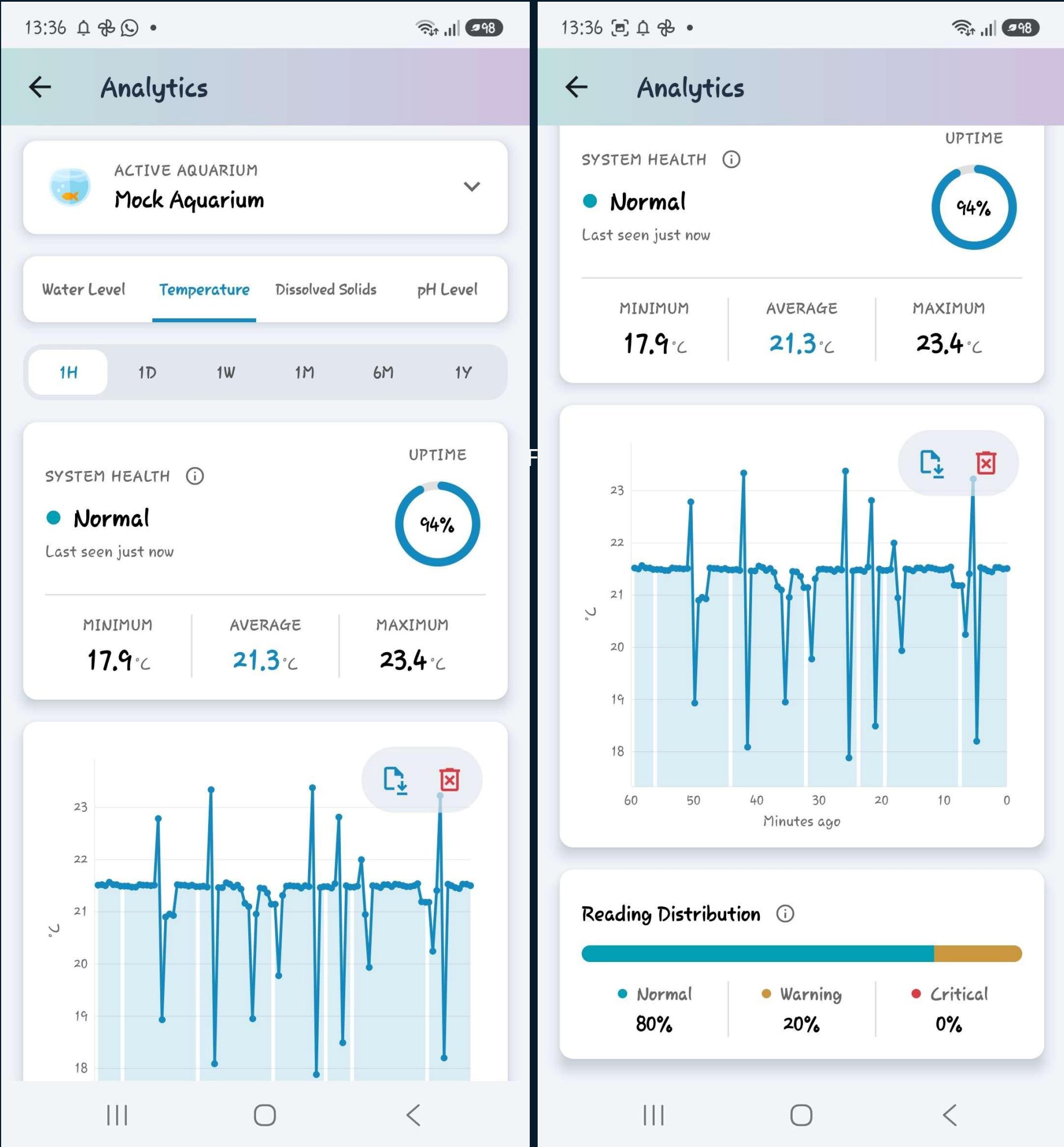


App Benefits: Analytics

From data collection to graphical analyses

Constant data is being pushed to the app. Simplifying the data into an easy to understand graph gives users the ability to comprehend their aquarium conditions.

NEW CHANGES → GRAPHICAL UPDATES



Associated Research

TYPES OF AQUARIUMS

Saltwater, freshwater, etc. tanks require different levels of pH, TDS, and temperature

AQUARIUM TEMPLATES

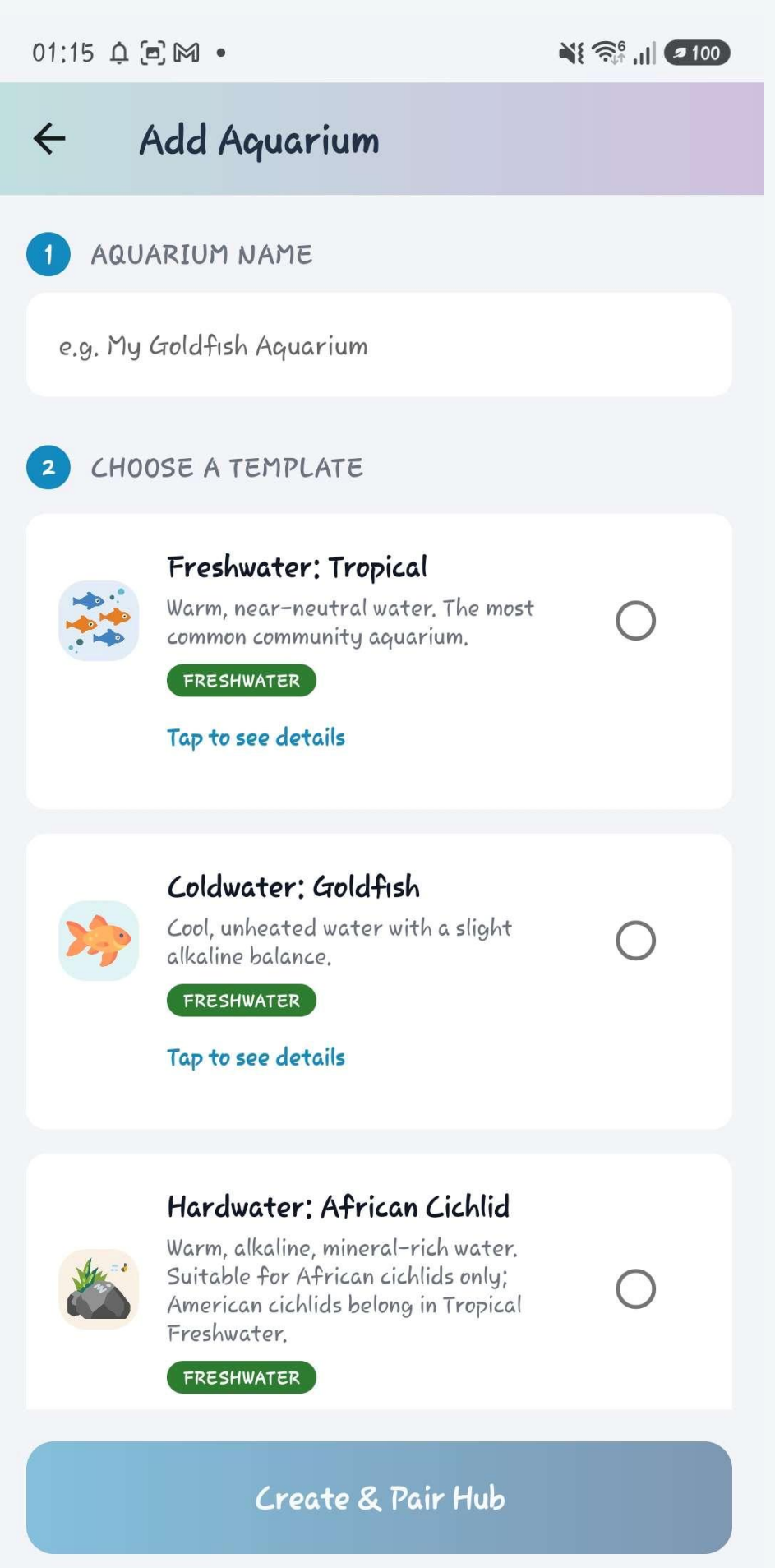
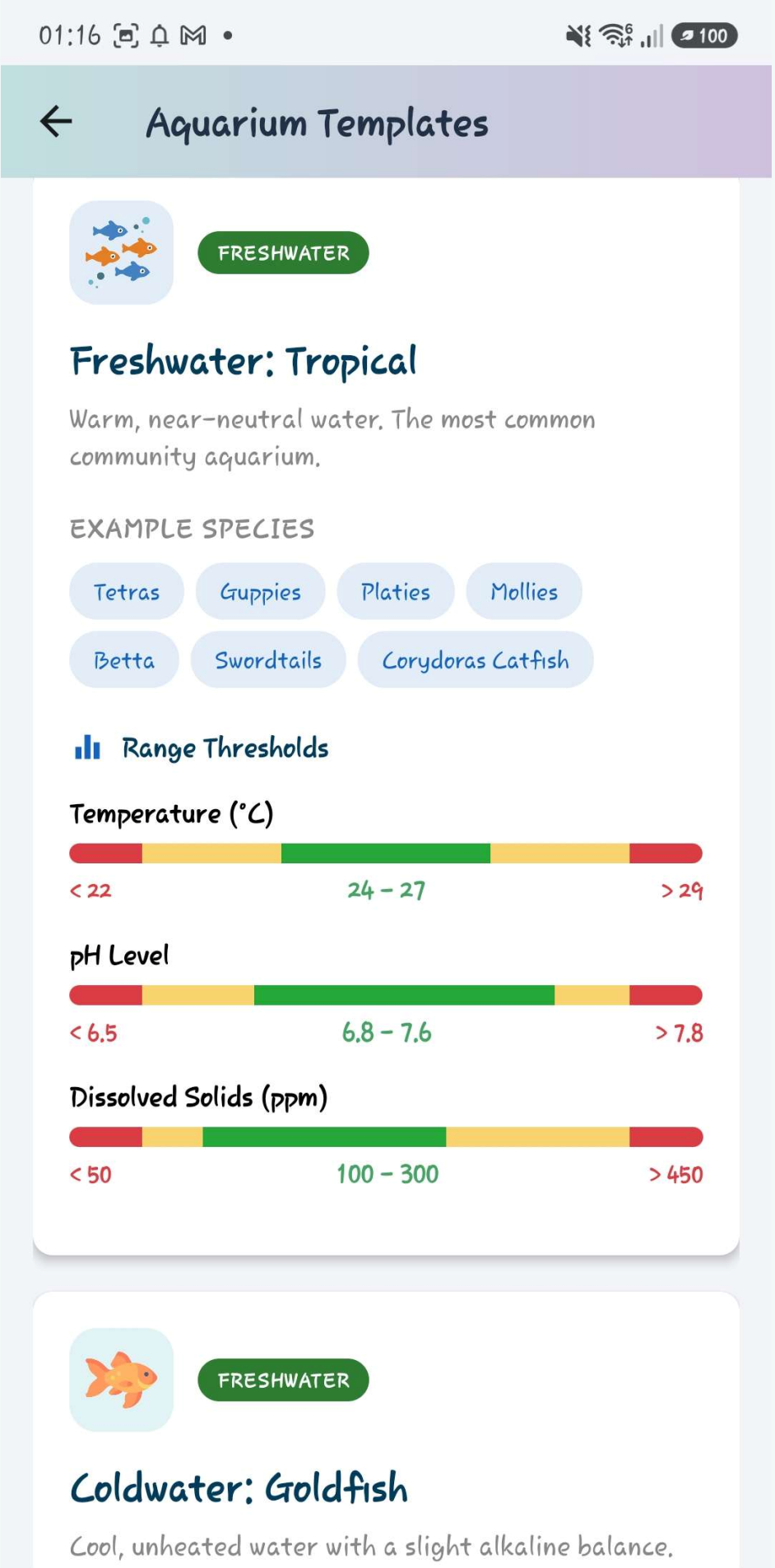
Our app features 4 built-in aquarium templates with set critical, warning, and safe range thresholds

RESEARCH REFERENCE

All of our aquarium template thresholds are based on extensive research referring to journal and conference papers and trusted aquarium guides & blogs

USER-CONFIGURABLE THRESHOLDS

Users can configure and set their own safe, warning and critical ranges as well as the abnormal spike duration



TRACTION & MILESTONES



Stakeholder Reviews

15

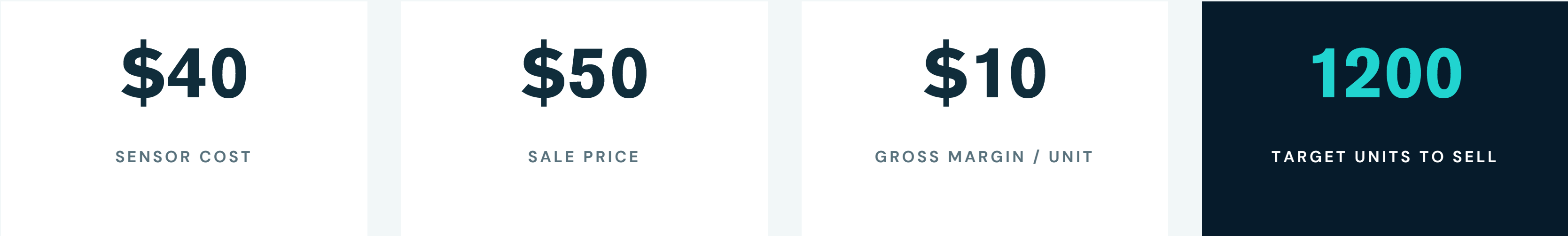
ROUNDS OF INTERVIEWS
WITH AQUARIUM OWNERS

**“The live updates keeps it very easy
to stay on top of what is going on in
each of my aquariums”**

AQUARIUM OWNER • CANADA

Overall positive feedback • Analytics praised for visual data representation

Overall Success



TEAM INVESTMENT

300 hours × \$40/hr

A reasonable price range relative to the long-term cost of aquariums and animal care.

OUR TEAM

A small team passionate about marine life:
Electrical and Computer engineering students focused on app design & IoT.



BILAL

Software Architect



DIPITA

Product Owner (Software)



DOMINIQUE

Hardware Bring-up



ARMAAN

Software Tester



NAVRAAJ

Structural Designer

AQUASENSE / INVESTMENT ASK

Join us in transforming aquarium care.

FUNDING USE

PRODUCT LAUNCH • MARKETING • R&D

LET'S BUILD SAFER AQUATIC ECOSYSTEMS

TEAM 6 • AQUARIUM INTELLIGENCE